

ShareGear

Final Consolidated Database Schema Design (v3.0)

Engine: SQL Server (T-SQL) · Generated natively, no ALTER TABLE statements

41 Tables Total · 12 Functional Parts

Executive Summary

This document — **ShareGear Database Schema Design v3.0** — fully supersedes v2.0 in its entirety. Version 3.0 integrates all Final Agreed GAP Decisions directly into the table definitions at their point of origin. Every accepted column, constraint, and index is declared inline on the table where it lives; no separate ALTER TABLE or CREATE INDEX pass is required to reach the hardened end state. Full-text indexes remain the one structural exception, placed immediately beneath their owning table.

Key Changes from v2.0

- **Tables Removed (team decision):** Offers (GAP-017 — IsNegotiable flag retained; counter-offer persistence out of scope), AuditLogs (GAP-026 — ledger/immutability hardening not applied).
- **Tables Added (10 new):** CustomerProfiles, CustomerTrustScoreHistory, ProviderTrustScoreHistory, BookingTimeAdjustmentRequests, ServiceListingAvailability, ConversationReadReceipts, SearchQueryLogs, DomainEventQueue, CategoryPricingAggregates, AiInteractionLogs.
- **Net Table Count:** 33 (v2.0) – 2 removed + 10 added = 41 tables.
- **Rejected GAPs (no schema impact):** GAP-006 (customer holdout election), GAP-013 (chat violation counter), GAP-003 priority-admin-gate portion. Note: the out-of-zone surcharge portion of GAP-003/004 IS applied.
- **Key Structural Hardening Applied:** GAP-005 native geography service zones + spatial indexes (ProviderProfiles, Bookings), GAP-021 high-water-mark read receipts (eliminates per-row lock contention), GAP-022 unique escrow constraints (double-payment prevention), GAP-024 / GAP-019 rowversion concurrency tokens (Bookings, EscrowRecords), GAP-028 full-text indexes (duplicate-listing detection).

Table of Contents

PART I — IDENTITY, ACCESS & ROLES

- Users
- RefreshTokens
- Addresses

PART II — PROVIDER, CUSTOMER & EMPLOYEE PROFILES

- ProviderProfiles
- ProviderTrustScoreHistory
- CustomerProfiles
- CustomerTrustScoreHistory
- EmployeeProfiles
- Operators

PART III — SERVICE CATALOG & LISTINGS

- Categories
- ServiceListings
- ServiceListingPhotos
- ServiceListingOperators
- ServiceListingAvailability
- BlackoutDates

PART IV — BOOKING LIFECYCLE & SCHEDULING

- Bookings
- BookingTimeAdjustmentRequests
- OperatorAssignments

PART V — MARKETPLACE & VERIFICATION DOCUMENTS

- MarketplaceListings
- MarketplaceListingPhotos
- MarketplaceOrders
- Documents

PART VI — UNIFIED ESCROW

- EscrowRecords

PART VII — REVIEWS & REPUTATION

- Reviews

PART VIII — CUSTOMER SERVICE & FIELD OPERATIONS

- Tickets
- TicketMessages
- TicketAttachments
- JobCompletionEvidenceForms
- JobCompletionEvidencePhotos
- FieldVerificationForms
- FieldVerificationPhotos

PART IX — REAL-TIME COMMUNICATION

- Conversations
- Messages
- ConversationReadReceipts
- Notifications

PART X — TRUST, RISK & OPERATIONAL LOGS

- ProviderIncidents

PART XI — AI, SEARCH & PLATFORM INTELLIGENCE

- SearchQueryLogs

DomainEventQueue

CategoryPricingAggregates

AIInteractionLogs

PART XII — PLATFORM CONFIGURATION

PlatformSettings

PART I — IDENTITY, ACCESS & ROLES

Users

Central identity store for every actor on the platform — Customers, Providers, Admins, and Employees. All other profile and activity tables reference this table as their root identity anchor.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Sequential GUID, primary key
Email	nvarchar(256)	NOT NULL	UNIQUE constraint	Login email address
PhoneNumber	nvarchar(20)	NOT NULL	UNIQUE constraint	Primary contact & OTP target
PasswordHash	nvarchar(255)	NOT NULL	—	Bcrypt-hashed credential
FirstName	nvarchar(100)	NOT NULL	—	Given name
LastName	nvarchar(100)	NOT NULL	—	Family name
Role	int	NOT NULL	CHECK 0–3; 0=Customer 1=Provider 2=Admin 3=Employee	Platform role discriminator
IsPhoneVerified	bit	NOT NULL	DEFAULT 0	SMS OTP verification flag
IsEmailVerified	bit	NOT NULL	DEFAULT 0	Email link verification flag
IsActive	bit	NOT NULL	DEFAULT 1	Soft-delete / suspension flag
FcmToken	nvarchar(500)	NULL	—	Firebase push token (optional)
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Account creation timestamp (UTC)
UpdatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Last profile update (UTC)

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- Foreign Keys: None
- **Index:** UQ_Users_Email — unique nonclustered on Email
- **Index:** UQ_Users_PhoneNumber — unique nonclustered on PhoneNumber

RefreshTokens

Stores JWT refresh tokens for every authenticated session, enabling secure token rotation and individual session revocation without invalidating all sessions for a user.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier

Column	Data Type	Null	Constraints / Defaults	Purpose
UserId	uniqueidentifier	NOT NULL	FK → Users(Id)	Owning user
Token	nvarchar(500)	NOT NULL	—	Opaque refresh token string
ExpiresAt	datetime2	NOT NULL	—	Token expiry timestamp (UTC)
IsRevoked	bit	NOT NULL	DEFAULT 0	Revocation flag
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Issuance timestamp (UTC)

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** UserId → Users(Id)
- **Index:** IX_RefreshTokens_UserId — nonclustered (UserId, IsRevoked)

Addresses

Stores one or more saved addresses per user, each with governorate/district hierarchy and precise decimal coordinates for distance calculations.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
UserId	uniqueidentifier	NOT NULL	FK → Users(Id)	Owning user
Label	nvarchar(100)	NULL	—	User-defined label (e.g. "Home")
Governorate	nvarchar(100)	NOT NULL	—	Top-level administrative area
District	nvarchar(100)	NOT NULL	—	Sub-district
Street	nvarchar(200)	NULL	—	Street-level detail (optional)
Latitude	decimal(10,7)	NOT NULL	—	7-decimal degree latitude
Longitude	decimal(10,7)	NOT NULL	—	7-decimal degree longitude
IsDefault	bit	NOT NULL	DEFAULT 0	Marks the user's primary address
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Creation timestamp (UTC)

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** UserId → Users(Id)
- **Index:** IX_Addresses_UserId — nonclustered (UserId)

PART II — PROVIDER, CUSTOMER & EMPLOYEE PROFILES

Includes GAP-005 (geography service-zone columns + spatial index), GAP-009 (customer trust score + history), GAP-010 (provider trust score + history). EmployeeProfiles carried forward unchanged from v2.0.

ProviderProfiles

Extends Users for the Provider role, storing business information, persisted trust score (GAP-010), service-zone geometry (GAP-005), search ranking modifier, and aggregate performance statistics.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
UserId	uniqueidentifier	NOT NULL	FK → Users(Id)	Linked user account
CompanyName	nvarchar(200)	NOT NULL	—	Trading / business name
BusinessDescription	nvarchar(1000)	NULL	—	Free-text company description
BaseLatitude	decimal(10,7)	NOT NULL	—	Operational base latitude
BaseLongitude	decimal(10,7)	NOT NULL	—	Operational base longitude
ServiceRadiusKm	int	NOT NULL	CHECK 0–50	Coverage radius in kilometres
OnboardingTier	int	NOT NULL	DEFAULT 0 · CHECK 0–3; 0=New 1=DocVerified 2=Established 3=Top	Verification tier
AverageRating	decimal(3,2)	NOT NULL	DEFAULT 0.00	Denormalised average review rating
TotalReviewsCount	int	NOT NULL	DEFAULT 0	Cumulative reviews count
CompletedBookingsCount	int	NOT NULL	DEFAULT 0	Cumulative completed bookings
ResponseRate	decimal(5,4)	NOT NULL	DEFAULT 0.0000	Ratio of bookings responded to in SLA
SearchRankingModifier	decimal(6,2)	NOT NULL	DEFAULT 0.00	Admin-adjustable ranking boost/penalty
TrustScore	decimal(5,2)	NOT NULL	DEFAULT 0.00 [GAP-010]	Persisted 5-component trust score
TrustLevel	int	NOT NULL	DEFAULT 1 · CHECK 0–3; 0=AtRisk 1=Standard 2=Verified 3=Top [GAP-010]	Banded trust level
TrustScoreLastComputedAt	datetime2	NULL	—	Last computation timestamp
ServiceZoneCenter	geography	NULL	[GAP-005] Spatial index SIX_ProviderProfiles_ServiceZoneCenter	Native geography point for STDistance()
ServiceZonePoly	geography	NULL	[GAP-005]	Polygon boundary for zone validation
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Profile creation timestamp
UpdatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Last modification timestamp

Special Elements & Hardened Features

- GAP-005 — Native geography columns (ServiceZoneCenter, ServiceZonePoly) replace decomposed lat/lng filtering. ServiceZoneCenter is backfilled post-deploy via a one-time UPDATE pipeline step.
- GAP-010 — TrustScore and TrustLevel are persisted to eliminate 5-component recomputation on every badge render.
- Spatial Index — SIX_ProviderProfiles_ServiceZoneCenter USING GEOGRAPHY_AUTO_GRID, CELLS_PER_OBJECT = 16.
- Partial Covering Index — IX_ProviderProfiles_AlternativeSearch on (ResponseRate DESC, AverageRating DESC, OnboardingTier DESC) WHERE ResponseRate >= 0.80, INCLUDE (UserId, CompanyName, BaseLatitude, BaseLongitude, ServiceRadiusKm, SearchRankingModifier, TrustLevel). Targets the Concierge Agent's GetAlternativeServices() path (GAP-030).

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** UserId → Users(Id)
- **Index:** SIX_ProviderProfiles_ServiceZoneCenter — SPATIAL GEOGRAPHY_AUTO_GRID (ServiceZoneCenter)
- **Index:** IX_ProviderProfiles_AlternativeSearch — partial covering (ResponseRate DESC, AverageRating DESC, OnboardingTier DESC) WHERE ResponseRate >= 0.80

ProviderTrustScoreHistory

Audit trail and trend-chart source for provider trust score changes (GAP-010). Each row captures a full score snapshot decomposed by weighted component, so drift in any single factor is independently auditable in the Section 8.1 admin view.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ProviderProfileId	uniqueidentifier	NOT NULL	FK → ProviderProfiles(Id)	Owning provider profile
TrustScore	decimal(5,2)	NOT NULL	—	Composite score snapshot
TrustLevel	int	NOT NULL	CHECK 0–3	Banded level snapshot
ComponentRating	decimal(5,2)	NULL	30% weight	Rating component (30%)
ComponentCompletion	decimal(5,2)	NULL	25% weight	Completion rate component (25%)
ComponentResponse	decimal(5,2)	NULL	20% weight	Response rate component (20%)
ComponentDocs	decimal(5,2)	NULL	15% weight	Documentation component (15%)
ComponentIncident	decimal(5,2)	NULL	-10% penalty	Incident penalty component (-10%)
TriggerEvent	nvarchar(50)	NULL	—	Event that triggered recomputation
RecordedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Snapshot timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ProviderProfileId → ProviderProfiles(Id)

- **Index:** IX_ProviderTrustHistory_Provider — nonclustered (ProviderProfileId, RecordedAt DESC)

CustomerProfiles

Symmetric trust-scoring extension for the demand side (GAP-009). Enables the Section 7.6 premium-service identity gate without a 4-table join on the booking-initiation hot path. One row per customer user (unique on UserId).

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
UserId	uniqueidentifier	NOT NULL	FK → Users(Id) · UNIQUE	Linked user (1:1)
TrustScore	decimal(5,2)	NOT NULL	DEFAULT 100.00	Composite customer trust score
CancellationRate	decimal(5,4)	NULL	—	Ratio of bookings cancelled
DisputeFrequencyScore	decimal(5,2)	NULL	—	Weighted dispute frequency
PaymentFailureCount	int	NOT NULL	DEFAULT 0	Cumulative payment failures
ReviewAuthenticityScore	decimal(5,2)	NULL	—	AI review authenticity metric
RequiresAdditionalVerification	bit	NOT NULL	DEFAULT 0	Premium-service identity gate flag
TotalBookings	int	NOT NULL	DEFAULT 0	Lifetime bookings count
TrustScoreLastComputedAt	datetime2	NULL	—	Last computation timestamp
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Profile creation timestamp
UpdatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Last modification timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** UserId → Users(Id)
- **Index:** UQ_CustomerProfiles_UserId — unique on UserId (enforces 1:1 with Users)

CustomerTrustScoreHistory

Audit trail for customer trust score changes. Each row records a score snapshot and the trigger event, supporting trend analysis and dispute investigation.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
CustomerProfileId	uniqueidentifier	NOT NULL	FK → CustomerProfiles(Id)	Owning customer profile
TrustScore	decimal(5,2)	NOT NULL	—	Score snapshot at this point

Column	Data Type	Null	Constraints / Defaults	Purpose
TriggerEvent	nvarchar(50)	NULL	—	Event that caused recomputation
RecordedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Snapshot timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** CustomerProfileId → CustomerProfiles(Id)
- **Index:** IX_CustomerTrustHistory_Profile — nonclustered (CustomerProfileId, RecordedAt DESC)

EmployeeProfiles

Extends Users for the Employee role, tracking department, assigned governorate, and dispatch availability for field verification operations. Carried forward unchanged from v2.0.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
UserId	uniqueidentifier	NOT NULL	FK → Users(Id) · UNIQUE	Linked user (1:1)
EmployeeCode	nvarchar(20)	NOT NULL	UNIQUE constraint	Internal employee identifier
Department	nvarchar(100)	NULL	—	Assigned department
AssignedGovernorate	nvarchar(100)	NULL	—	Operational region
IsAvailableForDispatch	bit	NOT NULL	DEFAULT 0	Real-time dispatch availability
TotalVerificationsCompleted	int	NOT NULL	DEFAULT 0	Lifetime completed verifications
TotalTicketsHandled	int	NOT NULL	DEFAULT 0	Lifetime ticket interactions
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Profile creation timestamp
UpdatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Last modification timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** UserId → Users(Id)
- **Index:** IX_EmployeeProfiles_Dispatch — nonclustered (IsAvailableForDispatch, AssignedGovernorate)

Operators

Individual human operators (equipment operators / technicians) belonging to a Provider, with licensing details. Assigned to service listings and bookings as the on-site resource.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier

Column	Data Type	Null	Constraints / Defaults	Purpose
ProviderProfileId	uniqueidentifier	NOT NULL	FK → ProviderProfiles(Id)	Owning provider
FullName	nvarchar(150)	NOT NULL	—	Operator's full name
YearsOfExperience	int	NOT NULL	DEFAULT 0	Declared years of experience
Specialization	nvarchar(150)	NULL	—	Area of expertise
LicenseType	nvarchar(50)	NULL	—	License category
LicenseNumber	nvarchar(100)	NULL	—	License reference number
LicenseExpiryDate	date	NULL	—	License expiry date
ProfilePhotoUrl	nvarchar(500)	NULL	—	Profile photo storage URL
IsActive	bit	NOT NULL	DEFAULT 1	Active / deactivated flag
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Record creation timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ProviderProfileId → ProviderProfiles(Id)
- **Index:** IX_Operators_ProviderProfileId — nonclustered (ProviderProfileId, IsActive)

PART III — SERVICE CATALOG & LISTINGS

Includes GAP-012 (structured *ServiceListingAvailability* replacing free-text *AvailabilityHours*), GAP-014 (*BlackoutDates* operator scoping), GAP-008 (Qdrant sync durability), GAP-011 (admin rejection fields), GAP-032 (quality-score breakdown), GAP-028 (full-text index for duplicate detection).

Categories

Self-referencing hierarchical taxonomy supporting both the Service and Marketplace category trees via the Type discriminator column.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	int (IDENTITY)	NOT NULL	PK · IDENTITY(1,1)	Auto-increment integer PK
Name	nvarchar(150)	NOT NULL	—	Display name
Slug	nvarchar(150)	NOT NULL	UNIQUE constraint	URL-safe slug
Type	int	NOT NULL	0=Service tree 1=Marketplace tree	Category tree discriminator
ParentId	int	NULL	FK → Categories(Id) self-ref	Parent category (NULL = root)

Relational Keys & Indexes

- **Primary Key:** Id (IDENTITY — integer)
- **FK:** ParentId → Categories(Id) (self-referencing)
- **Index:** UQ_Categories_Slug — unique on Slug

ServiceListings

Core service listing entity. Carries AI moderation flags (AiRiskScore, AiRiskLevel), persisted quality-score components (GAP-032), admin rejection audit fields (GAP-011), and Qdrant vector-sync state (GAP-008). Note: the legacy free-text *AvailabilityHours* column is removed; structured availability now lives in *ServiceListingAvailability*.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ProviderProfileId	uniqueidentifier	NOT NULL	FK → ProviderProfiles(Id)	Owning provider
CategoryId	int	NOT NULL	FK → Categories(Id)	Listing category
Title	nvarchar(200)	NOT NULL	Full-text indexed (GAP-028)	Listing headline
Description	nvarchar(max)	NOT NULL	Full-text indexed (GAP-028)	Detailed service description
Tags	nvarchar(500)	NULL	—	Comma-separated discovery tags
EquipmentModel	nvarchar(150)	NULL	—	Equipment model name
EquipmentCapacity	nvarchar(150)	NULL	—	Equipment capacity description

Column	Data Type	Null	Constraints / Defaults	Purpose
EquipmentCondition	int	NULL	0=Excellent 1=Good 2=Fair	Condition grading
YearOfManufacture	int	NULL	—	Equipment manufacture year
EquipmentRegistrationNumber	nvarchar(100)	NULL	—	Registration / serial number
HourlyRate	decimal(10,2)	NOT NULL	CHECK >= 0	Base hourly price
DailyRate	decimal(10,2)	NULL	—	Optional daily rate
MinimumBookingHours	int	NOT NULL	DEFAULT 1	Minimum booking duration (hrs)
Status	int	NOT NULL	DEFAULT 0; 0=Draft 1=PendingReview 2=Active 3=Rejected 4=Suspended 5=Inactive	Lifecycle status
AiRiskScore	int	NULL	CHECK 0–100	AI-computed moderation risk score
AiRiskLevel	nvarchar(20)	NULL	LOW MEDIUM HIGH	Banded AI risk level
AiRiskFlags	nvarchar(max)	NULL	—	Serialised AI flag details
AiRecommendation	nvarchar(50)	NULL	APPROVE APPROVE_WITH_CONDITIONS REJECT	AI disposition recommendation
QualityScore	int	NULL	CHECK 0–100	Composite listing quality score
QsDescription	int	NULL	CHECK 0–20 [GAP-032]	Description sub-score (max 20)
QsPhotos	int	NULL	CHECK 0–15 [GAP-032]	Photos sub-score (max 15)
QsSpecs	int	NULL	CHECK 0–15 [GAP-032]	Specs sub-score (max 15)
QsPricing	int	NULL	CHECK 0–10 [GAP-032]	Pricing sub-score (max 10)
QsOperator	int	NULL	CHECK 0–15 [GAP-032]	Operator sub-score (max 15)
QsDocs	int	NULL	CHECK 0–15 [GAP-032]	Documents sub-score (max 15)
QsZone	int	NULL	CHECK 0–10 [GAP-032]	Zone sub-score (max 10)
QualityScoreComputedAt	datetime2	NULL	—	Last quality score computation
AdminRejectionNote	nvarchar(500)	NULL	[GAP-011]	Admin-authored rejection reason
RejectedByAdminId	uniqueidentifier	NULL	FK → Users(Id) [GAP-011]	Admin who performed rejection
RejectedAt	datetime2	NULL	[GAP-011]	Rejection timestamp
SubmissionCount	int	NOT NULL	DEFAULT 1 [GAP-011]	Number of moderation submissions
QdrantPointId	nvarchar(100)	NULL	[GAP-008]	Vector DB record ID
LastEmbeddedAt	datetime2	NULL	[GAP-008]	Last successful embedding timestamp
EmbeddingStatus	int	NOT NULL	DEFAULT 0 · CHECK 0–3; 0=Pending 1=Synced 2=Failed 3=Deleted [GAP-008]	Qdrant sync state
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Listing creation timestamp
UpdatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Last modification timestamp

Special Elements & Hardened Features

- Full-Text Index (GAP-028) — ShareGearFTSCatalog on (Title, Description) with KEY INDEX PK_ServiceListings and STOPLIST = SYSTEM. Used by CheckDuplicateListing() in the ModeratorAgent.
- Quality Score Components (GAP-032) — Seven persisted sub-score columns (Qs*) eliminate 7 recomputation calls on the Concierge Agent's "Improve Your Listing" checklist path.
- Qdrant Sync State (GAP-008) — EmbeddingStatus + QdrantPointId + LastEmbeddedAt allow the DomainEventQueue to reconcile vector-DB drift after process restarts.
- Legacy Removal — AvailabilityHours (free-text) removed; structured per-day hours now in ServiceListingAvailability (GAP-012).

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ProviderProfileId → ProviderProfiles(Id)
- **FK:** CategoryId → Categories(Id)
- **FK:** RejectedByAdminId → Users(Id)
- **Index:** IX_ServiceListings_ProviderProfileId — nonclustered (ProviderProfileId, Status)
- **Index:** IX_ServiceListings_CategoryId — nonclustered (CategoryId, Status)
- **Index:** FULLTEXT on (Title, Description) via ShareGearFTSCatalog

ServiceListingPhotos

Stores ordered photo URLs for service listings. Display order is used for presentation sequencing in the UI.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ListingId	uniqueidentifier	NOT NULL	FK → ServiceListings(Id)	Owning listing
PhotoUrl	nvarchar(500)	NOT NULL	—	Blob storage URL
DisplayOrder	int	NOT NULL	DEFAULT 0	Sort order for UI rendering
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Upload timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ListingId → ServiceListings(Id)
- **Index:** IX_ServiceListingPhotos_ListingId_Order — nonclustered (ListingId, DisplayOrder)

ServiceListingOperators

Many-to-many junction linking service listings to the operators qualified to execute them. Used as the entry point for operator conflict-detection sub-queries (GAP-023).

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
ListingId	uniqueidentifier	NOT NULL	FK → ServiceListings(Id)	Listing side of the association
OperatorId	uniqueidentifier	NOT NULL	FK → Operators(Id)	Operator side of the association

Special Elements & Hardened Features

- Composite PK — (ListingId, OperatorId) — natural uniqueness enforced at the key level.
- Covering Index (GAP-023) — IX_ServiceListingOperators_ListingId on (ListingId) INCLUDE (OperatorId) — eliminates extra key lookups in the operator double-booking conflict query.

Relational Keys & Indexes

- **Primary Key:** (ListingId, OperatorId) — composite
- **FK:** ListingId → ServiceListings(Id)
- **FK:** OperatorId → Operators(Id)
- **Index:** IX_ServiceListingOperators_ListingId — nonclustered (ListingId) INCLUDE (OperatorId)

ServiceListingAvailability

Structured per-day operating hours for each listing (GAP-012), replacing the legacy unparseable AvailabilityHours text blob. Enables an indexed booking-time validation query instead of application-side string parsing.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ListingId	uniqueidentifier	NOT NULL	FK → ServiceListings(Id)	Owning listing
DayOfWeek	int	NOT NULL	CHECK 0–6; 0=Sunday ... 6=Saturday	Day of week
OpenTime	time	NOT NULL	—	Opening time
CloseTime	time	NOT NULL	CHECK CloseTime > OpenTime	Closing time

Special Elements & Hardened Features

- Business Rule — exactly one availability row per (ListingId, DayOfWeek), enforced by unique index UX_ListingAvailability_Day.
- CHECK constraint CK_ServiceListingAvailability_TimeRange ensures OpenTime < CloseTime at the schema level.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ListingId → ServiceListings(Id)
- **Index:** UX_ListingAvailability_Day — UNIQUE (ListingId, DayOfWeek)

BlackoutDates

Records dates on which a specific service listing — or a specific operator assigned to it — is unavailable. Used in the combined conflict-detection predicate alongside OperatorAssignments (GAP-014).

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ListingId	uniqueidentifier	NOT NULL	FK → ServiceListings(Id)	Affected listing

Column	Data Type	Null	Constraints / Defaults	Purpose
OperatorId	uniqueidentifier	NULL	FK → Operators(Id) [GAP-014]	Operator-scoped blackout (NULL = listing-wide)
[Date]	date	NOT NULL	—	Blacked-out calendar date
Reason	nvarchar(300)	NULL	—	Optional reason text
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Record creation timestamp

Special Elements & Hardened Features

- GAP-014 — OperatorId column allows operator-level blackout scoping in addition to listing-level, wiring into the full conflict-detection predicate set alongside OperatorAssignments.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ListingId → ServiceListings(Id)
- **FK:** OperatorId → Operators(Id)
- **Index:** IX_BlackoutDates_ListingId_Date — nonclustered (ListingId, [Date])

PART IV — BOOKING LIFECYCLE & SCHEDULING

Includes GAP-001 (cancellation persistence; emergency-override fields excluded per team decision), GAP-002 (time-adjustment requests), GAP-003/004 (out-of-zone surcharge only; IsPriorityBooking removed), GAP-005 (ServiceLocationGeo), GAP-015 (reassignment linkage), GAP-023 (conflict-detection index), GAP-024 (rowversion concurrency token).

Bookings

Central booking record tracking the full lifecycle of a service engagement from draft through completion or dispute resolution. Contains location, scheduling, financial snapshot, cancellation bands, surcharge decomposition, and reassignment audit linkage.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
CustomerId	uniqueidentifier	NOT NULL	FK → Users(Id)	Booking customer
ServiceListingId	uniqueidentifier	NOT NULL	FK → ServiceListings(Id)	Service being booked
AssignedOperatorId	uniqueidentifier	NULL	FK → Operators(Id)	Assigned on-site operator
JobTitle	nvarchar(150)	NOT NULL	—	Customer-provided job label
JobDescription	nvarchar(1000)	NULL	—	Detailed job requirements
Governorate	nvarchar(100)	NOT NULL	—	Site governorate
District	nvarchar(100)	NOT NULL	—	Site district
Street	nvarchar(200)	NULL	—	Site street detail
Latitude	decimal(10,7)	NULL	—	Site latitude
Longitude	decimal(10,7)	NULL	—	Site longitude
ServiceLocationGeo	geography	NULL	[GAP-005]	Native geography point for zone validation
SiteContactName	nvarchar(150)	NOT NULL	—	On-site contact name
SiteContactPhone	nvarchar(20)	NOT NULL	—	On-site contact phone
AccessRequirements	nvarchar(500)	NULL	—	Site access notes
SafetyNotes	nvarchar(500)	NULL	—	Site safety notes
RequestedStartDate	date	NOT NULL	—	Requested service date
RequestedStartTime	time	NOT NULL	—	Requested start time
EstimatedDurationHours	decimal(5,2)	NOT NULL	CHECK > 0	Initial estimated duration
ActualDurationHours	decimal(5,2)	NULL	—	Final recorded duration
HourlyRateSnapshot	decimal(10,2)	NOT NULL	—	Rate at booking time (immutable)
EstimatedTotal	decimal(10,2)	NOT NULL	CHECK >= 0	Calculated estimated cost
IsOutOfZoneBooking	bit	NOT NULL	DEFAULT 0 [GAP-003/004]	Zone boundary flag

Column	Data Type	Null	Constraints / Defaults	Purpose
OutOfZoneDistanceKm	decimal(8,2)	NULL	[GAP-003/004]	Distance beyond provider zone (km)
OutOfZoneSurchargeAmount	decimal(10,2)	NULL	[GAP-003/004]	Out-of-zone surcharge value
OutOfZoneSurchargeAcceptedAt	datetime2	NULL	[GAP-003/004]	Customer surcharge acceptance timestamp
PrioritySurchargeAmount	decimal(10,2)	NULL	—	Priority booking surcharge
FuelSurchargeAmount	decimal(10,2)	NULL	—	Fuel surcharge
SurchargeAmount	decimal (COMPUTED)	—	PERSISTED = Ooz + Priority + Fuel	Total surcharge (computed, persisted)
Status	int	NOT NULL	DEFAULT 0 · CHECK 0–15 (see enum below)	Booking lifecycle status
ProviderRejectionReason	nvarchar(300)	NULL	—	Provider rejection note
CancellationReason	nvarchar(300)	NULL	—	Cancellation reason text
CancelledAt	datetime2	NULL	[GAP-001]	Cancellation timestamp
CancellationInitiatedByRole	int	NULL	CHECK 0–2; 0=Customer 1=Provider 2=Admin [GAP-001]	Who initiated cancellation
CancellationRefundPct	decimal(5,2)	NULL	[GAP-001]	Refund percentage applied (from policy bands)
ProviderCancellationPenaltyApplied	bit	NOT NULL	DEFAULT 0	Provider penalty flag
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Booking creation timestamp
ConfirmedAt	datetime2	NULL	—	Provider acceptance timestamp
PaymentCapturedAt	datetime2	NULL	—	Payment capture timestamp
StartedAt	datetime2	NULL	—	Job start timestamp
CompletedMarkedAt	datetime2	NULL	—	Provider completion mark timestamp
CompletionConfirmedAt	datetime2	NULL	—	Customer confirmation timestamp
DisputeOpenedAt	datetime2	NULL	—	Dispute open timestamp
FieldVerificationDispatchedAt	datetime2	NULL	—	Field verification dispatch timestamp
ReassignedToBookingId	uniqueidentifier	NULL	FK → Bookings(Id) [GAP-015]	New booking created on reassignment
OriginalBookingId	uniqueidentifier	NULL	FK → Bookings(Id) [GAP-015]	Original booking this was created from
ReassignedAt	datetime2	NULL	[GAP-015]	Reassignment timestamp
[Timestamp]	rowversion	NOT NULL	[GAP-024] optimistic concurrency token	Auto-updated binary concurrency stamp

Special Elements & Hardened Features

- Status Enum (0–15) — 0=Draft, 1=PendingProviderResponse, 2=ConfirmedPendingPayment, 3=Active, 4=InProgress, 5=PendingCustomerConfirmation, 6=Completed, 7=Disputed, 8=Rejected, 9=ProviderUnresponsive, 10=Reassigned, 11=CancelledRefunded, 12=ResolvedReleased, 13=ResolvedRefunded, 14=PendingFieldVerification, 15=FieldVerificationComplete.

- rowversion [Timestamp] (GAP-024) — Optimistic-concurrency token guarding against simultaneous writers (provider-accept vs SLA watchdog) corrupting the booking state machine.
- SurchargeAmount — Computed persisted column = ISNULL(OOZ,0) + ISNULL(Priority,0) + ISNULL(Fuel,0). Never written directly.
- Reassignment Self-FK (GAP-015) — CK_Bookings_NoSelfReassign prevents Id = ReassignedToBookingId or Id = OriginalBookingId.
- GAP-001 Exclusions — EmergencyCancellation, EmergencyOverrideStatus, EmergencyOverrideAdminId, EmergencyOverrideDecidedAt were explicitly excluded by team decision.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** CustomerId → Users(Id)
- **FK:** ServiceListingId → ServiceListings(Id)
- **FK:** AssignedOperatorId → Operators(Id)
- **FK:** ReassignedToBookingId → Bookings(Id) [self-ref]
- **FK:** OriginalBookingId → Bookings(Id) [self-ref]
- **Index:** IX_Bookings_CustomerId — nonclustered (CustomerId, Status)
- **Index:** IX_Bookings_ServiceListingId — nonclustered (ServiceListingId, Status)
- **Index:** IX_Bookings_AssignedOperatorId — nonclustered (AssignedOperatorId, RequestedStartDate)

BookingTimeAdjustmentRequests

Structured approval workflow for a provider's request to extend a booking by up to 2 hours above the original estimate (GAP-002). Replaces direct unaudited mutation of Bookings.EstimatedDurationHours.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
BookingId	uniqueidentifier	NOT NULL	FK → Bookings(Id)	Target booking
RequestedAdditionalHrs	decimal(4,2)	NOT NULL	CHECK > 0 AND <= 2.0	Additional hours requested (max 2)
AdditionalCostAmount	decimal(10,2)	NOT NULL	CHECK >= 0	Cost of additional hours
Status	int	NOT NULL	DEFAULT 0 · CHECK 0–2; 0=Pending 1=Accepted 2=Rejected	Approval status
ProviderNote	nvarchar(300)	NULL	—	Provider's justification note
CustomerAcknowledgedAt	datetime2	NULL	—	Customer acknowledgement timestamp
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Request creation timestamp

Special Elements & Hardened Features

- Unique Pending Guard — UX_TimeAdj_BookingId_Pending: UNIQUE (BookingId) WHERE Status = 0 — only one pending adjustment per booking at any time.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** BookingId → Bookings(Id)
- **Index:** UX_TimeAdj_BookingId_Pending — unique filtered (BookingId) WHERE Status = 0

- **Index:** IX_TimeAdj_BookingId — nonclustered (BookingId, Status)

OperatorAssignments

Records the confirmed schedule window for an operator on a specific booking, enabling the partial index-based conflict-detection query (GAP-023) used at both submission time and provider-acceptance time.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
OperatorId	uniqueidentifier	NOT NULL	FK → Operators(Id)	Assigned operator
BookingId	uniqueidentifier	NOT NULL	FK → Bookings(Id)	Associated booking
ScheduledStart	datetime2	NOT NULL	—	Assignment start datetime
ScheduledEnd	datetime2	NOT NULL	CHECK ScheduledEnd > ScheduledStart	Assignment end datetime
Status	int	NOT NULL	3=Active 4=InProgress (booking-status mirrors)	Schedule-blocking status
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Record creation timestamp

Special Elements & Hardened Features

- **Conflict-Detection Index (GAP-023)** — IX_OperatorAssignments_ConflictDetection: nonclustered partial index on (OperatorId, Status, ScheduledStart, ScheduledEnd) INCLUDE (BookingId) WHERE Status IN (3, 4). Combined with BlackoutDates (GAP-014) forms the complete double-booking predicate set.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** OperatorId → Operators(Id)
- **FK:** BookingId → Bookings(Id)
- **Index:** IX_OperatorAssignments_ConflictDetection — partial covering (OperatorId, Status, ScheduledStart, ScheduledEnd) INCLUDE (BookingId) WHERE Status IN (3, 4)

PART V — MARKETPLACE & VERIFICATION DOCUMENTS

Includes GAP-016 (MarketplaceOrders cancellation fields), GAP-011 (admin rejection), GAP-008 (Qdrant sync), GAP-028 (full-text). GAP-017 removes the Offers table — IsNegotiable flag is retained on listings but round-based counter-offer persistence is out of scope.

MarketplaceListings

Equipment and goods listings for the Marketplace module, mirroring the moderation and Qdrant sync pattern of ServiceListings. Carries AI risk flags (GAP-008), admin rejection context (GAP-011), and full-text search support (GAP-028).

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
SellerId	uniqueidentifier	NOT NULL	FK → Users(Id)	Listing owner
CategoryId	int	NOT NULL	FK → Categories(Id)	Marketplace category
Title	nvarchar(200)	NOT NULL	Full-text indexed	Listing headline
Condition	int	NOT NULL	0=New 1=LikeNew 2=Good 3=Fair 4=ForPartsOnly	Item condition
YearOfManufacture	int	NULL	—	Item manufacture year
Description	nvarchar(max)	NOT NULL	Full-text indexed	Detailed item description
Specifications	nvarchar(max)	NULL	—	Technical specifications
Price	decimal(12,2)	NOT NULL	CHECK >= 0	Listed price
IsNegotiable	bit	NOT NULL	DEFAULT 0	Price negotiable flag
TransactionMethod	int	NOT NULL	0=BuyerCollects 1=SellerDelivers 2=ShippingOnly	Fulfilment method
Governorate	nvarchar(100)	NOT NULL	—	Seller's governorate
District	nvarchar(100)	NULL	—	Seller's district
Status	int	NOT NULL	DEFAULT 0; 0=Draft 1=PendingReview 2=Active 3=Sold 4=Removed 5=Rejected	Lifecycle status
AiRiskScore	int	NULL	CHECK 0–100	AI moderation risk score
AiRiskLevel	nvarchar(20)	NULL	LOW MEDIUM HIGH	Banded AI risk level
AiRiskFlags	nvarchar(max)	NULL	—	AI flag detail payload
VideoUrl	nvarchar(500)	NULL	—	Optional video showcase URL
AdminRejectionNote	nvarchar(500)	NULL	[GAP-011]	Admin rejection reason
RejectedByAdminId	uniqueidentifier	NULL	FK → Users(Id) [GAP-011]	Rejecting admin
RejectedAt	datetime2	NULL	[GAP-011]	Rejection timestamp
SubmissionCount	int	NOT NULL	DEFAULT 1 [GAP-011]	Moderation submission count
QdrantPointId	nvarchar(100)	NULL	[GAP-008]	Qdrant vector record ID

Column	Data Type	Null	Constraints / Defaults	Purpose
LastEmbeddedAt	datetime2	NULL	[GAP-008]	Last embedding timestamp
EmbeddingStatus	int	NOT NULL	DEFAULT 0 · CHECK 0–3 [GAP-008]	Vector sync state
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Creation timestamp
UpdatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Last update timestamp

Special Elements & Hardened Features

- Full-Text Index (GAP-028) — On (Title, Description) via ShareGearFTSCatalog, STOPLIST = SYSTEM.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** SellerId → Users(Id)
- **FK:** CategoryId → Categories(Id)
- **FK:** RejectedByAdminId → Users(Id)
- **Index:** IX_MarketplaceListings_SellerId — nonclustered (SellerId, Status)
- **Index:** IX_MarketplaceListings_CategoryId — nonclustered (CategoryId, Status)
- **Index:** FULLTEXT on (Title, Description)

MarketplaceListingPhotos

Ordered photo gallery for Marketplace listings, following the same pattern as ServiceListingPhotos.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ListingId	uniqueidentifier	NOT NULL	FK → MarketplaceListings(Id)	Owning listing
PhotoUrl	nvarchar(500)	NOT NULL	—	Storage URL
DisplayOrder	int	NOT NULL	DEFAULT 0	Presentation order
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Upload timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ListingId → MarketplaceListings(Id)
- **Index:** IX_MarketplaceListingPhotos_ListingId_Order — nonclustered (ListingId, DisplayOrder)

MarketplaceOrders

Purchase record for a marketplace item, with explicit cancellation state (GAP-016) distinguishing pre-dispatch from post-dispatch cancellation to support the refund-policy branch defined in Section 7.8.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier

Column	Data Type	Null	Constraints / Defaults	Purpose
BuyerId	uniqueidentifier	NOT NULL	FK → Users(Id)	Purchasing user
ListingId	uniqueidentifier	NOT NULL	FK → MarketplaceListings(Id)	Item purchased
Amount	decimal(12,2)	NOT NULL	CHECK >= 0	Order amount
DeliveryAddress	nvarchar(500)	NULL	—	Delivery address (if applicable)
DeliveryPreference	int	NOT NULL	0=BuyerCollects 1=SellerDelivers 2=ShippingOnly	Fulfilment method
TrackingNumber	nvarchar(100)	NULL	—	Shipment tracking reference
Status	int	NOT NULL	DEFAULT 0 · CHECK 0–9 [GAP-016]	Order lifecycle status
SellerConfirmedAt	datetime2	NULL	—	Seller confirmation timestamp
DispatchedAt	datetime2	NULL	—	Dispatch timestamp
DeliveredAt	datetime2	NULL	—	Delivery confirmation timestamp
ConfirmedByBuyerAt	datetime2	NULL	—	Buyer receipt confirmation
CancellationReason	nvarchar(300)	NULL	[GAP-016]	Cancellation reason
CancelledAt	datetime2	NULL	[GAP-016]	Cancellation timestamp
CancellationInitiatedByRole	int	NULL	CHECK 0–2; 0=Buyer 1=Seller 2=Admin [GAP-016]	Cancellation initiator
ReturnShippingCost	decimal(10,2)	NULL	[GAP-016]	Return shipping cost (post-dispatch)
ReturnShippingAcceptedByBuyerAt	datetime2	NULL	[GAP-016]	Buyer return-cost acceptance timestamp
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Order creation timestamp

Special Elements & Hardened Features

- Status Enum (GAP-016) — 0=PendingPayment, 1=PaymentCaptured, 2=SellerConfirmed, 3=Dispatched, 4=Delivered, 5=Completed, 6=Disputed, 7=CancelledPreDispatch, 8=CancelledPostDispatch, 9=Refunded.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** BuyerId → Users(Id)
- **FK:** ListingId → MarketplaceListings(Id)
- **Index:** IX_MarketplaceOrders_BuyerId — nonclustered (BuyerId, Status)
- **Index:** IX_MarketplaceOrders_ListingId — nonclustered (ListingId, Status)

Documents

Centralised document repository used for equipment certificates, operator licenses, business registration, and insurance files. Polymorphic FK design allows attachment to a User, ServiceListing, MarketplaceListing, or Operator.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
UserId	uniqueidentifier	NOT NULL	FK → Users(Id)	Uploading user
ServiceListingId	uniqueidentifier	NULL	FK → ServiceListings(Id)	Associated service listing (optional)
MarketplaceListingId	uniqueidentifier	NULL	FK → MarketplaceListings(Id)	Associated marketplace listing (optional)
OperatorId	uniqueidentifier	NULL	FK → Operators(Id)	Associated operator (optional)
DocumentType	int	NOT NULL	EquipmentCertificate OperatorLicense BusinessRegistration Insurance	Document category
FileUrl	nvarchar(500)	NOT NULL	—	Blob storage URL
Status	int	NOT NULL	Queued Processing Verified Unreadable Expired	AI processing status
ExtractedText	nvarchar(max)	NULL	—	OCR-extracted document text
ConfidenceScore	decimal(5,4)	NULL	CHECK 0–1	AI extraction confidence (0.0–1.0)
KeyFieldsPresent	bit	NULL	—	AI field-presence check result
ExpiryDate	date	NULL	—	Document expiry date
ExpiryStatus	int	NULL	Valid Expired Unreadable	Derived expiry classification
FailureReason	nvarchar(500)	NULL	—	AI processing failure description
UploadedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Upload timestamp
VerifiedAt	datetime2	NULL	—	Verification completion timestamp

Special Elements & Hardened Features

- Filtered indexes on ServiceListingId and MarketplaceListingId include WHERE IS NOT NULL, so scans are only against rows with an actual FK value.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** UserId → Users(Id)
- **FK:** ServiceListingId → ServiceListings(Id)
- **FK:** MarketplaceListingId → MarketplaceListings(Id)
- **FK:** OperatorId → Operators(Id)
- **Index:** IX_Documents_UserId — nonclustered (UserId)
- **Index:** IX_Documents_ServiceListingId — filtered nonclustered (ServiceListingId) WHERE NOT NULL
- **Index:** IX_Documents_MarketplaceListingId — filtered nonclustered (MarketplaceListingId) WHERE NOT NULL

PART VI — UNIFIED ESCROW

Includes GAP-002 (adjustment-request linkage), GAP-018 (commission-rate snapshot), GAP-019 (rowversion + earliest-release timestamp), GAP-022 (unique constraints preventing duplicate escrow rows).

EscrowRecords

Holds payment funds in escrow for both Booking and Marketplace Order transactions. Unique filtered indexes (GAP-022) provide the structural defence against duplicate-row double-payment; the rowversion token (GAP-019) prevents the double-release race condition.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
BookingId	uniqueidentifier	NULL	FK → Bookings(Id) [GAP-022]	Associated booking (mutually exclusive with Order)
MarketplaceOrderId	uniqueidentifier	NULL	FK → MarketplaceOrders(Id) [GAP-022]	Associated order (mutually exclusive with Booking)
GrossAmount	decimal(12,2)	NOT NULL	CHECK >= 0	Total captured amount
PlatformCommission	decimal(12,2)	NOT NULL	CHECK >= 0	Platform fee retained
ProviderPayout	decimal(12,2)	NOT NULL	CHECK >= 0	Amount released to provider
VatAmount	decimal(12,2)	NULL	—	VAT component
PartialSettleCustomerAmount	decimal(12,2)	NULL	—	Customer partial settlement amount
PartialSettleProviderAmt	decimal(12,2)	NULL	—	Provider partial settlement amount
Status	int	NOT NULL	AwaitingCapture Captured Held ReleasedToProvider RefundedToCustomer PartiallySettled	Escrow lifecycle status
PaymentGatewayReference	nvarchar(200)	NULL	—	Payment gateway transaction ID
AdditionalHoldDays	int	NULL	DEFAULT 0 · CHECK >= 0	Admin-extended hold period (days)
CapturedAt	datetime2	NULL	—	Capture timestamp
HeldAt	datetime2	NULL	—	Hold start timestamp
ReleasedAt	datetime2	NULL	—	Release timestamp
FrozenAt	datetime2	NULL	—	Freeze timestamp (dispute)
FreezeReason	nvarchar(300)	NULL	—	Reason for freeze
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Record creation timestamp
AdjustmentRequestId	uniqueidentifier	NULL	FK → BookingTimeAdjustmentRequests(Id) [GAP-002]	Links escrow change to approved time-extension
CommissionRateSnapshot	decimal(5,4)	NOT NULL	DEFAULT 0.1000 [GAP-018]	Commission rate at capture time (audit trail)
[Timestamp]	rowversion	NOT NULL	[GAP-019] optimistic concurrency	Concurrency token (auto-updated)
EarliestReleaseAt	datetime2	NULL	[GAP-019] fraud-prevention cool-down	Earliest eligible payout release timestamp

Special Elements & Hardened Features

- GAP-022 — UX_EscrowRecords_BookingId: UNIQUE (BookingId) WHERE BookingId IS NOT NULL and UX_EscrowRecords_OrderId: UNIQUE (MarketplaceOrderId) WHERE MarketplaceOrderId IS NOT NULL — structural defence against retry-loop duplicate inserts enabling double payout.
- GAP-019 — rowversion [Timestamp] prevents the double-release race condition documented in Section 9.3.
- GAP-018 — CommissionRateSnapshot makes every historical PlatformCommission independently verifiable against the rate that produced it.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** BookingId → Bookings(Id)
- **FK:** MarketplaceOrderId → MarketplaceOrders(Id)
- **FK:** AdjustmentRequestId → BookingTimeAdjustmentRequests(Id)
- **Index:** UX_EscrowRecords_BookingId — unique filtered (BookingId) WHERE NOT NULL
- **Index:** UX_EscrowRecords_OrderId — unique filtered (MarketplaceOrderId) WHERE NOT NULL

PART VII — REVIEWS & REPUTATION

Includes GAP-029 (direct listing FKs eliminating a 3-table join on the review-moderation hot path).

Reviews

Stores post-transaction reviews for both booking and marketplace order contexts. Direct listing FK columns (GAP-029) allow the ModeratorAgent's authenticity check to be a single-row read. A check constraint enforces mutually exclusive context (booking vs. marketplace).

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ReviewerId	uniqueidentifier	NOT NULL	FK → Users(Id)	Author of the review
ReviewedUserId	uniqueidentifier	NOT NULL	FK → Users(Id)	Subject of the review
BookingId	uniqueidentifier	NULL	FK → Bookings(Id)	Associated booking (exclusive with marketplace)
MarketplaceOrderId	uniqueidentifier	NULL	FK → MarketplaceOrders(Id)	Associated order (exclusive with booking)
ServiceListingId	uniqueidentifier	NULL	FK → ServiceListings(Id) [GAP-029]	Direct listing context (service side)
MarketplaceListingId	uniqueidentifier	NULL	FK → MarketplaceListings(Id) [GAP-029]	Direct listing context (marketplace side)
Rating	int	NOT NULL	CHECK 1–5	Star rating
Comment	nvarchar(2000)	NULL	—	Review text
ModerationStatus	int	NOT NULL	Clean FlaggedWarning Blocked	AI moderation status
IsPublished	bit	NOT NULL	DEFAULT 0	Visibility flag
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Submission timestamp
PublishedAt	datetime2	NULL	—	Publication timestamp

Special Elements & Hardened Features

- **CK_Reviews_SingleContext** — Enforces that a review must belong to exactly one context: either (BookingId IS NOT NULL AND ServiceListingId IS NOT NULL) or (MarketplaceOrderId IS NOT NULL AND MarketplaceListingId IS NOT NULL) — never both, never neither.
- **GAP-029** — ServiceListingId and MarketplaceListingId direct FKs eliminate the Reviews → Bookings → ServiceListings → Categories join chain previously on the moderation hot path.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ReviewerId → Users(Id)
- **FK:** ReviewedUserId → Users(Id)
- **FK:** BookingId → Bookings(Id)
- **FK:** MarketplaceOrderId → MarketplaceOrders(Id)
- **FK:** ServiceListingId → ServiceListings(Id)
- **FK:** MarketplaceListingId → MarketplaceListings(Id)
- **Index:** IX_Reviews_ReviewedUserId — nonclustered (ReviewedUserId, IsPublished)

- **Index:** IX_Reviews_ServiceListingId — filtered (ServiceListingId) WHERE NOT NULL
- **Index:** IX_Reviews_MarketplaceListingId — filtered (MarketplaceListingId) WHERE NOT NULL

PART VIII — CUSTOMER SERVICE & FIELD OPERATIONS

Carried forward from v2.0 Sections 6.2–6.8, with GAP-020 (admin dispute SLA deadline) added to Tickets. All original v2.0 Appendix A indexes preserved and embedded inline.

Tickets

Support ticket lifecycle, covering all categories (service quality, payment, fraud, safety, marketplace) with AI-assisted triage fields, admin SLA tracking (GAP-020), and field-verification escalation linkage.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
TicketNumber	nvarchar(20)	NOT NULL	UNIQUE constraint	Human-readable ticket reference
SubmittedById	uniqueidentifier	NOT NULL	FK → Users(Id)	User who opened the ticket
AssignedToUserId	uniqueidentifier	NULL	FK → Users(Id)	Assigned admin/employee
ResolvedByUserId	uniqueidentifier	NULL	FK → Users(Id)	User who resolved the ticket
BookingId	uniqueidentifier	NULL	FK → Bookings(Id)	Associated booking (optional)
MarketplaceOrderId	uniqueidentifier	NULL	FK → MarketplaceOrders(Id)	Associated order (optional; exclusive with Booking)
Subject	nvarchar(200)	NOT NULL	—	Ticket subject line
Category	int	NOT NULL	CHECK 0–7; 0=ServiceQuality 1=OperatorConduct 2=PaymentIssue 3=Fraud 4=SafetyConcern 5=CompletionDispute 6=MarketplaceIssue 7=Other	Ticket category
Priority	int	NOT NULL	DEFAULT 0 · CHECK 0–2; 0=Normal 1=High 2=Urgent	Priority level
Status	int	NOT NULL	DEFAULT 0 · CHECK 0–7	Ticket lifecycle status
AiSummary	nvarchar(max)	NULL	—	AI-generated complaint summary
AiIdentifiedIssue	nvarchar(500)	NULL	—	AI-identified issue label
AiClaimedImpact	nvarchar(500)	NULL	—	AI-assessed impact description
AiEscalationPriority	int	NULL	CHECK 0–2	AI-recommended escalation priority
ResolutionType	int	NULL	CHECK 0–5; 0=FullRefund 1=FullRelease 2=Partial 3=Redo 4=NoAction 5=EscalatedToField	Final resolution category
AdminResolution	nvarchar(max)	NULL	—	Admin resolution notes
FirstResponseAt	datetime2	NULL	—	First admin response timestamp
ResolvedAt	datetime2	NULL	—	Resolution timestamp
ClosedAt	datetime2	NULL	—	Closure timestamp
ReopenedAt	datetime2	NULL	—	Reopen timestamp

Column	Data Type	Null	Constraints / Defaults	Purpose
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Submission timestamp
UpdatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Last update timestamp
AdminResponseDeadline	datetime2	NULL	[GAP-020]	SLA deadline (3 business days)
EscalatedAt	datetime2	NULL	[GAP-020]	Escalation timestamp
EscalatedToAdminId	uniqueidentifier	NULL	FK → Users(Id) [GAP-020]	Senior admin escalation target

Special Elements & Hardened Features

- Status Enum (0–7) — 0=Open, 1=InProgress, 2=PendingCustomerReply, 3=PendingProviderReply, 4=PendingFieldVerification, 5=Resolved, 6=Closed, 7=Reopened.
- SLA Index (GAP-020) — IX_Tickets_SlaWatch: nonclustered (AdminResponseDeadline, Status) WHERE Status IN (0, 1) AND AdminResponseDeadline IS NOT NULL — drives the "3 business days → escalate" watchdog.
- Unique Active Ticket Guard — UX_Tickets_BookingId_Active: UNIQUE (BookingId) WHERE BookingId IS NOT NULL AND Status <> 6 — enforces the v2.0 business rule of one active ticket per booking.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** SubmittedById → Users(Id)
- **FK:** AssignedToUserId → Users(Id)
- **FK:** ResolvedByUserId → Users(Id)
- **FK:** BookingId → Bookings(Id)
- **FK:** MarketplaceOrderId → MarketplaceOrders(Id)
- **FK:** EscalatedToAdminId → Users(Id)
- **Index:** UX_Tickets_BookingId_Active — unique filtered (BookingId) WHERE NOT NULL AND Status <> 6
- **Index:** IX_Tickets_AssignedToUserId_Status — nonclustered (AssignedToUserId, Status, Priority DESC, CreatedAt)
- **Index:** IX_Tickets_SlaWatch — nonclustered (AdminResponseDeadline, Status) WHERE Status IN (0,1) AND AdminResponseDeadline IS NOT NULL

TicketMessages

Threaded message log for a support ticket, supporting customer, provider, agent, and system message types with an internal flag for admin-only messages.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
TicketId	uniqueidentifier	NOT NULL	FK → Tickets(Id)	Owning ticket
SenderId	uniqueidentifier	NOT NULL	FK → Users(Id)	Message author
Body	nvarchar(3000)	NOT NULL	Hardened to 3,000-char app limit	Message body
MessageType	int	NOT NULL	CHECK 0–3; 0=Customer 1=Provider 2=Agent 3=System	Message type discriminator
IsInternal	bit	NOT NULL	DEFAULT 0	Internal admin-only flag

Column	Data Type	Null	Constraints / Defaults	Purpose
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Message send timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** TicketId → Tickets(Id)
- **FK:** SenderId → Users(Id)
- **Index:** IX_TicketMessages_TicketId_CreatedAt — nonclustered (TicketId, CreatedAt)
- **Index:** IX_TicketMessages_TicketId_IsInternal — nonclustered (TicketId, IsInternal, CreatedAt)

TicketAttachments

File attachments scoped to individual ticket messages, supporting images, documents, and video evidence.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
TicketMessageId	uniqueidentifier	NOT NULL	FK → TicketMessages(Id)	Owning message
UploadedByUserId	uniqueidentifier	NOT NULL	FK → Users(Id)	Uploader
FileUrl	nvarchar(500)	NOT NULL	—	Storage URL
FileName	nvarchar(255)	NOT NULL	—	Original filename
FileType	int	NOT NULL	CHECK 0–2; 0=Image 1=Document 2=Video	File category
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Upload timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** TicketMessageId → TicketMessages(Id)
- **FK:** UploadedByUserId → Users(Id)
- **Index:** IX_TicketAttachments_TicketMessageId — nonclustered (TicketMessageId)

JobCompletionEvidenceForms

Provider-submitted evidence of job completion, reviewed by an admin before escrow release. Exactly one form per booking enforced by a unique index.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
BookingId	uniqueidentifier	NOT NULL	FK → Bookings(Id)	Associated booking
SubmittedByUserId	uniqueidentifier	NOT NULL	FK → Users(Id)	Submitting provider/operator
ReviewedByAdminId	uniqueidentifier	NULL	FK → Users(Id)	Reviewing admin

Column	Data Type	Null	Constraints / Defaults	Purpose
ProviderNotes	nvarchar(1000)	NULL	—	Provider completion notes
Status	int	NOT NULL	DEFAULT 1 · CHECK 0–4; 0=Draft 1=Submitted 2=UnderAdminReview 3=AcceptedAsEvidence 4=RejectedAsEvidence	Review status
AdminReviewNote	nvarchar(500)	NULL	—	Admin review comment
SubmittedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Submission timestamp
ReviewedAt	datetime2	NULL	—	Admin review timestamp
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Creation timestamp

Special Elements & Hardened Features

- **UX_JobCompletionEvidenceForms_BookingId** — UNIQUE (BookingId) — enforces exactly one evidence form per booking.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** BookingId → Bookings(Id)
- **FK:** SubmittedByUserId → Users(Id)
- **FK:** ReviewedByAdminId → Users(Id)
- **Index:** UX_JobCompletionEvidenceForms_BookingId — unique (BookingId)

JobCompletionEvidencePhotos

Ordered photo evidence submitted by providers in a job completion evidence form.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
FormId	uniqueidentifier	NOT NULL	FK → JobCompletionEvidenceForms(Id)	Owning form
PhotoUrl	nvarchar(500)	NOT NULL	—	Storage URL
Caption	nvarchar(200)	NULL	—	Optional photo caption
DisplayOrder	int	NOT NULL	DEFAULT 0	Presentation order
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Upload timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** FormId → JobCompletionEvidenceForms(Id)
- **Index:** IX_JobCompletionEvidencePhotos_FormId_Order — nonclustered (FormId, DisplayOrder)

FieldVerificationForms

Records the complete lifecycle of an on-site field verification visit — from dispatch through employee visit, AI photo similarity analysis, and final admin decision.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
BookingId	uniqueidentifier	NOT NULL	FK → Bookings(Id)	Associated booking
DispatchedEmployeeId	uniqueidentifier	NOT NULL	FK → Users(Id)	Dispatched employee
DispatchedByAdminId	uniqueidentifier	NOT NULL	FK → Users(Id)	Dispatching admin
DecidedByAdminId	uniqueidentifier	NULL	FK → Users(Id)	Admin who made the final decision
LinkedEvidenceFormId	uniqueidentifier	NOT NULL	FK → JobCompletionEvidenceForms(Id)	Linked provider evidence
DispatchInstructions	nvarchar(1000)	NULL	—	Admin dispatch instructions
VisitStatus	int	NOT NULL	DEFAULT 0 · CHECK 0–3; 0=Dispatched 1=OnSite 2=Completed 3=FailedAccess	Visit progress status
EmployeeNotes	nvarchar(2000)	NULL	—	Employee field notes
FieldVerificationOutcome	int	NULL	CHECK 0–3; 0=JobConfirmed 1=NotConfirmed 2=PartiallyConfirmed 3=Inconclusive	Employee outcome assessment
AiSimilarityScore	decimal(5,4)	NULL	CHECK 0–1	AI photo similarity score (0.0–1.0)
AiSimilarityNotes	nvarchar(1000)	NULL	—	AI comparison notes
AdminDecision	int	NOT NULL	DEFAULT 0 · CHECK 0–3; 0=Pending 1=ReleaseToProvider 2=RefundToCustomer 3=PartialSettlement	Final admin decision
AdminDecisionNote	nvarchar(500)	NULL	—	Admin decision rationale
DispatchedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Dispatch timestamp
VisitedAt	datetime2	NULL	—	On-site arrival timestamp
FormSubmittedAt	datetime2	NULL	—	Employee form submission timestamp
DecidedAt	datetime2	NULL	—	Admin decision timestamp
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Record creation timestamp

Special Elements & Hardened Features

- AdminDecision sequencing guard is enforced in the service layer (AdminDecision may only be set once VisitStatus = 2 Completed and AiSimilarityScore IS NOT NULL) — not at schema level.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** BookingId → Bookings(Id)
- **FK:** DispatchedEmployeeId → Users(Id)
- **FK:** DispatchedByAdminId → Users(Id)
- **FK:** DecidedByAdminId → Users(Id)
- **FK:** LinkedEvidenceFormId → JobCompletionEvidenceForms(Id)
- **Index:** UX_FieldVerificationForms_BookingId — unique (BookingId)

- **Index:** IX_FieldVerificationForms_AdminDecision_Status — nonclustered (AdminDecision, VisitStatus, DispatchedAt)

FieldVerificationPhotos

Photos taken by the dispatched employee during the on-site field verification visit, used as input for AI similarity analysis against the provider's job completion evidence.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
FieldVerificationFormId	uniqueidentifier	NOT NULL	FK → FieldVerificationForms(Id)	Owning form
PhotoUrl	nvarchar(500)	NOT NULL	—	Storage URL
Caption	nvarchar(200)	NULL	—	Optional caption
DisplayOrder	int	NOT NULL	DEFAULT 0	Presentation order
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Upload timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** FieldVerificationFormId → FieldVerificationForms(Id)
- **Index:** IX_FieldVerificationPhotos_FormId_Order — nonclustered (FieldVerificationFormId, DisplayOrder)

PART IX — REAL-TIME COMMUNICATION

Includes GAP-021 (ConversationReadReceipts — high-water-mark pattern replaces per-message IsRead/ReadAt mutation, eliminating page-level lock contention) and GAP-025 (Notifications covering indexes).

Conversations

Represents a messaging thread between two users (initiator and participant), optionally anchored to a service listing, marketplace listing, or booking. A unique filtered index prevents duplicate active threads for the same listing/pair.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ServiceListingId	uniqueidentifier	NULL	FK → ServiceListings(Id)	Anchoring service listing (optional)
MarketplaceListingId	uniqueidentifier	NULL	FK → MarketplaceListings(Id)	Anchoring marketplace listing (optional)
BookingId	uniqueidentifier	NULL	FK → Bookings(Id)	Anchoring booking (optional)
InitiatedById	uniqueidentifier	NOT NULL	FK → Users(Id)	Conversation initiator
ParticipantId	uniqueidentifier	NOT NULL	FK → Users(Id)	Other participant
IsLocked	bit	NOT NULL	DEFAULT 0	Locked (no new messages allowed)
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Thread creation timestamp
LockedAt	datetime2	NULL	—	Lock timestamp

Special Elements & Hardened Features

- **Unique Thread Guard** — UX_Conversations_ServiceListing_Participants: UNIQUE (ServiceListingId, InitiatedById, ParticipantId) WHERE ServiceListingId IS NOT NULL — prevents a customer from opening a second active thread with the same provider for the same listing.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ServiceListingId → ServiceListings(Id)
- **FK:** MarketplaceListingId → MarketplaceListings(Id)
- **FK:** BookingId → Bookings(Id)
- **FK:** InitiatedById → Users(Id)
- **FK:** ParticipantId → Users(Id)
- **Index:** UX_Conversations_ServiceListing_Participants — unique filtered (ServiceListingId, InitiatedById, ParticipantId) WHERE NOT NULL
- **Index:** IX_Conversations_ParticipantId — nonclustered (ParticipantId, IsLocked)

Messages

Individual messages within a conversation thread. Note: IsRead and ReadAt are intentionally absent (removed by GAP-021); read state is maintained in ConversationReadReceipts as a high-water mark to eliminate per-row UPDATE lock contention.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ConversationId	uniqueidentifier	NOT NULL	FK → Conversations(Id)	Owning conversation
SenderId	uniqueidentifier	NOT NULL	FK → Users(Id)	Message author
Content	nvarchar(1000)	NOT NULL	Hardened to 1,000-char app limit	Message text
MessageType	int	NOT NULL	0=Text 1=Image 2=System	Message content type
AttachmentUrl	nvarchar(500)	NULL	—	Attachment storage URL (Image type)
IsBlocked	bit	NOT NULL	DEFAULT 0	Moderation block flag
SentAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Send timestamp

Special Elements & Hardened Features

- GAP-021 — IsRead and ReadAt removed entirely. Unread count queries compare Messages.SentAt against ConversationReadReceipts.LastReadAt (high-water mark) instead of scanning/updating individual rows.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ConversationId → Conversations(Id)
- **FK:** SenderId → Users(Id)
- **Index:** IX_Messages_ConversationId_SentAt — nonclustered (ConversationId, SentAt)

ConversationReadReceipts

Per-user, per-conversation high-water mark for read state (GAP-021). Replaces per-message IsRead mutation with a single upsert on LastReadAt and LastReadMessageId, eliminating the page-level lock contention pattern under concurrent chat load.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
ConversationId	uniqueidentifier	NOT NULL	FK → Conversations(Id) · part of PK	Conversation
UserId	uniqueidentifier	NOT NULL	FK → Users(Id) · part of PK	Reading user
LastReadMessageId	uniqueidentifier	NULL	FK → Messages(Id)	Last read message pointer
LastReadAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	High-water-mark timestamp

Special Elements & Hardened Features

- Composite PK on (ConversationId, UserId) — one receipt row per user per conversation.
- Unread-count pattern: SELECT COUNT(*) FROM Messages WHERE ConversationId = @id AND SenderId <> @userId AND SentAt > (SELECT ISNULL(LastReadAt, "1900-01-01") FROM ConversationReadReceipts WHERE ConversationId = @id AND UserId = @userId).

Relational Keys & Indexes

- **Primary Key:** (ConversationId, UserId) — composite
- **FK:** ConversationId → Conversations(Id)

- **FK:** UserId → Users(Id)
- **FK:** LastReadMessageId → Messages(Id)
- **Indexes:** None beyond PK clustered index

Notifications

Persistent notification records across In-App, Push, and SMS channels. Covering indexes (GAP-025) optimise the notification bell's primary read pattern and the admin notification queue.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
UserId	uniqueidentifier	NOT NULL	FK → Users(Id)	Recipient user
EventType	nvarchar(100)	NOT NULL	—	Notification event identifier
Title	nvarchar(200)	NOT NULL	—	Notification title
Body	nvarchar(1000)	NULL	—	Notification body text
ReferenceId	nvarchar(100)	NULL	—	Entity ID the notification relates to
ReferenceType	nvarchar(50)	NULL	—	Entity type (e.g. "Booking")
IsRead	bit	NOT NULL	DEFAULT 0	Read flag
Channel	int	NOT NULL	0=InApp 1=Push 2=SMS	Delivery channel
SentAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Sent timestamp
ReadAt	datetime2	NULL	—	Read timestamp

Special Elements & Hardened Features

- **Covering Index (GAP-025)** — IX_Notifications_UserUnread: (UserId, IsRead, SentAt DESC) INCLUDE (EventType, Title, Body, ReferenceId, ReferenceType, Channel) — eliminates key-lookup amplification for the notification bell query.
- **Admin Queue Index (GAP-025)** — IX_Notifications_AdminQueue: (Channel, IsRead, SentAt DESC) WHERE IsRead = 0.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** UserId → Users(Id)
- **Index:** IX_Notifications_UserUnread — covering (UserId, IsRead, SentAt DESC) INCLUDE (all display cols)
- **Index:** IX_Notifications_AdminQueue — filtered (Channel, IsRead, SentAt DESC) WHERE IsRead = 0

PART X — TRUST, RISK & OPERATIONAL LOGS

GAP-026 decision: AuditLogs table intentionally removed from v3.0. The team opted not to apply ledger/immutability hardening. ProviderIncidents is the sole table in this part.

ProviderIncidents

Logs negative incidents against a provider (unresponsive bookings, cancellation penalties) that feed into the ComponentIncident weight of the provider trust score computation.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
ProviderProfileId	uniqueidentifier	NOT NULL	FK → ProviderProfiles(Id)	Affected provider
BookingId	uniqueidentifier	NULL	FK → Bookings(Id)	Associated booking (optional)
IncidentType	int	NOT NULL	e.g. UnresponsiveIncident CancellationPenalty	Incident category
PenaltyApplied	bit	NOT NULL	DEFAULT 0	Whether a trust score penalty was applied
AdminNote	nvarchar(500)	NULL	—	Admin context note
OccurredAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Incident timestamp

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** ProviderProfileId → ProviderProfiles(Id)
- **FK:** BookingId → Bookings(Id)
- **Index:** IX_ProviderIncidents_ProviderProfileId — nonclustered (ProviderProfileId, OccurredAt DESC)

PART XI — AI, SEARCH & PLATFORM INTELLIGENCE

New cross-cutting infrastructure: GAP-007 (search analytics), GAP-008 (durable Qdrant sync queue), GAP-027 (pre-aggregated category pricing), GAP-031 (AI interaction / false-positive feedback log).

SearchQueryLogs

Persists every search query and its outcome metadata (GAP-007) for the Section 8.3 semantic-vs-keyword pattern metric, zero-result rate analysis, and Concierge Agent performance monitoring.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
UserId	uniqueidentifier	NULL	FK → Users(Id)	Searching user (NULL = anonymous)
SessionId	nvarchar(100)	NULL	—	Client session identifier
RawQuery	nvarchar(1000)	NOT NULL	—	Original query string
ExtractedIntentJson	nvarchar(max)	NULL	—	AI-parsed intent as JSON
ContextDomain	int	NOT NULL	0=Services 1=Marketplace 2=All	Search domain context
SearchMode	int	NOT NULL	DEFAULT 0; 0=Semantic 1=Keyword	Search mode used
ResultCount	int	NOT NULL	—	Number of results returned
HasLowConfidence	bit	NOT NULL	DEFAULT 0	Low-confidence result flag
HasZeroResults	bit	NOT NULL	DEFAULT 0	Zero-results flag
AlternativeSuggested	bit	NOT NULL	DEFAULT 0	Whether an alternative was offered
AlternativeAcceptedAt	datetime2	NULL	—	Alternative acceptance timestamp
ProcessingMs	int	NULL	—	Query processing time (ms)
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Query timestamp

Special Elements & Hardened Features

- Covering analytics index — IX_SearchQueryLogs_CreatedAt: (CreatedAt DESC) INCLUDE (ContextDomain, SearchMode, ResultCount, HasZeroResults, AlternativeSuggested, AlternativeAcceptedAt, ProcessingMs) — supports dashboard aggregation without key lookups.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** UserId → Users(Id)
- **Index:** IX_SearchQueryLogs_CreatedAt — covering (CreatedAt DESC) INCLUDE (analytics cols)
- **Index:** IX_SearchQueryLogs_UserId — nonclustered (UserId, CreatedAt DESC) WHERE UserId IS NOT NULL

DomainEventQueue

Durable outbox for listing state-change events that must be reflected in Qdrant (GAP-008). Prevents event loss on process restart — the prior in-memory bus provided no such guarantee. A retry counter and failure reason support dead-letter handling.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
EventType	nvarchar(100)	NOT NULL	ListingApproved ListingEdited ListingDeleted ListingQualityScoreStale	Event category
EntityType	nvarchar(50)	NOT NULL	ServiceListing MarketplaceListing	Affected entity type
EntityId	uniqueidentifier	NOT NULL	—	Affected entity ID
Status	int	NOT NULL	DEFAULT 0; 0=Pending 1=Processing 2=Done 3=Failed	Processing status
RetryCount	int	NOT NULL	DEFAULT 0	Number of processing attempts
FailureReason	nvarchar(500)	NULL	—	Last failure detail
ProcessedAt	datetime2	NULL	—	Completion timestamp
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Enqueue timestamp

Special Elements & Hardened Features

- Processing Index — IX_DomainEventQueue_Processing: (Status, CreatedAt) WHERE Status IN (0, 3) — targets only Pending and Failed rows for the background processor query.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- Foreign Keys: None
- **Index:** IX_DomainEventQueue_Processing — filtered nonclustered (Status, CreatedAt) WHERE Status IN (0, 3)

CategoryPricingAggregates

Pre-aggregated pricing statistics per category, governorate, and price type (GAP-027). Refreshed every 30 minutes by a background IHostedService. Both CheckPriceAnomaly() (60-second SLA) and the customer price-fairness badge read this table directly instead of running live PERCENTILE_CONT() window functions against ServiceListings.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
CategoryId	int	NOT NULL	FK → Categories(Id) · part of PK	Category
LocationGovernorate	nvarchar(100)	NOT NULL	Part of PK	Geographic filter
PriceType	int	NOT NULL	0=Hourly 1=Daily · Part of PK	Price type
MedianPrice	decimal(10,2)	NOT NULL	—	Median price in this segment
Percentile25	decimal(10,2)	NOT NULL	—	25th percentile price

Column	Data Type	Null	Constraints / Defaults	Purpose
Percentile75	decimal(10,2)	NOT NULL	—	75th percentile price
MinPrice	decimal(10,2)	NOT NULL	—	Minimum observed price
MaxPrice	decimal(10,2)	NOT NULL	—	Maximum observed price
SampleCount	int	NOT NULL	—	Active listings in this segment
ComputedAt	datetime2	NOT NULL	—	Aggregation timestamp

Special Elements & Hardened Features

- Composite PK on (CategoryId, LocationGovernorate, PriceType) — no surrogate key; the natural key fully identifies a pricing segment.

Relational Keys & Indexes

- **Primary Key:** (CategoryId, LocationGovernorate, PriceType) — composite natural PK
- **FK:** CategoryId → Categories(Id)
- Indexes: None beyond PK clustered index

AiInteractionLogs

Correlates every AI agent recommendation with any subsequent admin override (GAP-031), providing the data required to compute the false-positive rate metric for the Section 8.2 AI Risk Dashboard. A computed persisted ExpiresAt column anchors the 90-day retention policy from Section 9.4.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
Id	uniqueidentifier	NOT NULL	PK · DEFAULT NEWSEQUENTIALID()	Row identifier
AgentType	int	NOT NULL	0=ConciergeAgent 1=ModeratorAgent	AI agent that produced the recommendation
InvocationContext	nvarchar(100)	NOT NULL	e.g. ListingModeration ReviewFilter Search ComplaintSummarization	Invocation scenario
EntityType	nvarchar(50)	NULL	—	Entity type acted upon
EntityId	uniqueidentifier	NULL	—	Entity ID acted upon
AiRecommendation	nvarchar(50)	NULL	APPROVE REJECT FLAG	AI output recommendation
AiRiskScore	int	NULL	CHECK 0–100	AI risk score at invocation
AdminOverride	nvarchar(50)	NULL	—	Admin's actual decision (if different)
AdminOverrideById	uniqueidentifier	NULL	FK → Users(Id)	Admin who overrode the AI
AdminOverrideAt	datetime2	NULL	—	Override timestamp
LatencyMs	int	NULL	—	AI response latency (ms)
InputTokens	int	NULL	—	Tokens consumed (input)
OutputTokens	int	NULL	—	Tokens consumed (output)
CreatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Log creation timestamp
ExpiresAt	datetime2 (COMPUTED)	—	PERSISTED = DATEADD(day,90,CreatedAt)	90-day retention expiry (computed, persisted)

Special Elements & Hardened Features

- **ExpiresAt** — Computed persisted column (DATEADD(day, 90, CreatedAt)) anchoring the 90-day retention policy; the expiry index enables efficient purge jobs.
- **False-Positive Index** — IX_AiInteractionLogs_FalsePositive: (AiRecommendation, AdminOverride, CreatedAt DESC) WHERE both are NOT NULL — powers the false-positive rate computation for the AI Risk Dashboard.

Relational Keys & Indexes

- **Primary Key:** Id (NEWSEQUENTIALID)
- **FK:** AdminOverrideById → Users(Id)
- **Index:** IX_AiInteractionLogs_EntityId — nonclustered (EntityId, AgentType) WHERE NOT NULL
- **Index:** IX_AiInteractionLogs_FalsePositive — filtered (AiRecommendation, AdminOverride, CreatedAt DESC) WHERE both NOT NULL
- **Index:** IX_AiInteractionLogs_Expiry — nonclustered (ExpiresAt)

PART XII — PLATFORM CONFIGURATION

PlatformSettings

Key-value configuration store for runtime-tunable platform parameters (SLA windows, commission rates, price-anomaly thresholds, cancellation policy bands). Seeded at deployment; mutated only by admin users via authenticated API endpoints.

Field Definitions

Column	Data Type	Null	Constraints / Defaults	Purpose
SettingKey	nvarchar(100)	NOT NULL	PK (natural key)	Unique setting identifier
SettingValue	nvarchar(max)	NOT NULL	—	Setting value (string or JSON)
Description	nvarchar(500)	NULL	—	Human-readable description of the setting
UpdatedByAdminId	uniqueidentifier	NULL	FK → Users(Id)	Last modifying admin
UpdatedAt	datetime2	NOT NULL	DEFAULT GETUTCDATE()	Last modification timestamp

Special Elements & Hardened Features

- Seed values included at deployment (non-DDL DML): FieldVerificationDispatchSlaHours (4h), FieldVerificationCompletionSlaHours (24h), ProviderResponseSlaHours (2h), CustomerAutoConfirmHours (4h), MarketplaceAutoConfirmHours (72h), AdminDisputeResolutionSlaDays (3), SemanticSearchSimilarityThreshold (0.65), PriceAnomalyLowPct (40%), PriceAnomalyHighPct (200%), CommissionRate (0.10), DuplicateListingThreshold (70), CancellationPolicyBands (JSON).

Relational Keys & Indexes

- **Primary Key:** SettingKey (natural string PK)
- **FK:** UpdatedByAdminId → Users(Id)
- **Indexes:** None beyond PK clustered index

ShareGear — Final Consolidated Database Schema Design v3.0 · 41 Tables · SQL Server (T-SQL)
This document supersedes Database Design v2.0 in full. All GAP decisions integrated inline.